Exp= 4.1

→

WAP to read a list of integers and store it in a single dimensional array. Write a C program to print the second largest integer in a list of arrays

⟶

```c
# include <stdio.h>

int main ()
{
  int n, i;
  printf (" enter number of elements :");
  scanf ("%d", &n);

  int arr [n];
  printf (" Enter %d integers :\n", n);
  for (i = 0; i < n; i++)
  {
     scanf ("%d", & arr [i]);
  }
```

Teacher's Signature

```
int largest, second_largest;

if (arr[0] > arr[1])
  {
    largest = arr[0]
    second_largest = arr[1];
  }
else
  {
    largest = arr[0];
    second_largest = arr[1];
  }

for (i = 2; i < n; i++)
 {
   if (arr[i] > largest)
    {
     second_largest = largest;
      largest = arr[i]
    }
   else if (arr[i] > second_largest && arr[i] < largest)
    {
     second_largest = arr[i];
    }
 }
```

```
printf (" The second largest element is : %d\n",
        second_largest );

return 0;
}
```

```c
#include <stdio.h>

int main()
{
    int n, i;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    int arr[n];

    printf("Enter %d integers:\n", n);
    for (i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    int largest, second_largest;

    if (arr[0] > arr[1])
    {
        largest = arr[0];
        second_largest = arr[1];
    }
    else
    {
        largest = arr[1];
        second_largest = arr[0];
    }

    for (i = 2; i < n; i++)
    {
        if (arr[i] > largest)
        {
            second_largest = largest;
            largest = arr[i];
        }
        else if (arr[i] > second_largest && arr[i] < largest)
        {
            second_largest = arr[i];
        }
    }

    printf("The second largest element is: %d\n", second_largest);

    return 0;
}
```

```
PS C:\Users\sanch\OneDrive\Desktop\c programming> cd 'c:\Users\sanch\OneDrive\Desktop\c programming\exp_4.c\output'
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_4.c\output> & .\'exp1.exe'
Enter the number of elements: 5
Enter 5 integers:
1  30 27 42 19
The second largest element is: 30
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_4.c\output>
```

Exp = 4.2

→

WAP to read a list of integers and store it in a single dimensional array. Write a C program to count and display positive, negative, odd and even numbers in an array

```
# include <stdio.h>

int main () {
 int n, i, positive, negative, zero ;
    positive = 0 ;
    negative = 0;
    zero = 0;

printf (" Enter the number of elements : ") ;
scanf ( " %d " , &n);

int arr [n]

printf (" Enter %d elements :", n);
for (i = 0 ; i < n ; i++ )
   {
     scanf ("%d", &n);
   }
```

```
for (i=0; i<n; i++)
  {
    if (arr (i] >0)
        positive ++;
    else if (arr [i] <0)
        negative ++;
    else
        zero ++;
  }

printf(" elements which are positive = %d\n", positive);
printf(" elements which are negative = %d\n", negative);
printf(" elements which are zero = %d\n", zero);

return 0;

}
```

Teacher's Signature ________

exp_4.c > C exp2.c > main()

```c
#include <stdio.h>

int main()
{
    int n, i, positive , negative , zero ;
    positive = 0;
    negative = 0;
    zero = 0;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    int arr[n];

    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    for (i = 0; i < n; i++)
    {
        if (arr[i] > 0)
            positive++;
        else if (arr[i] < 0)
            negative++;
        else
            zero++;
    }

    printf("elements which are positive = %d\n" , positive);
    printf("elements which are negative = %d\n" , negative);
    printf("elements which are zero = %d\n" , zero);

    return 0;
}
```

```
PS C:\Users\sanch\OneDrive\Desktop\c programming> cd 'c:\Users\sanch\OneDrive\Desktop\c programming\exp_4.c\output'
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_4.c\output> & .\'exp2.exe'

Enter the number of elements: 6
Enter 6 elements: 1 36 0 -8 -52 6
elements which are positive = 3
elements which are negative = 2
elements which are zero = 1
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_4.c\output>
```

Exp = 4.3

→

WAP to read a list of integers and store it in a single dimensional array. Write a C program to find the frequency of a particular number in a list of integers

→

```
# include < stdio.h >

int main ()
{
 int n, i, num, count = 0;

printf(" enter the number of elements : ");
scanf("%d", &n);

 int arr [n];

printf (" Enter %d integers : \n", n);
 for (i = 0; i <n; i++)
   {
```

Teacher's Signature

```
scanf ("%d", &arr[i]);
}

printf ("Enter the number to find its frequency:");
scanf ("%d", &num);

for (i=0 ; i< n ; i++ )
  {
     if (arr[i] == num )
       {
          count ++;
       }
  }

printf ("The frequency of %d is %d \n", num, count);

return 0;

}
```

xp_4.c > C exp3.c > main()

```c
#include <stdio.h>

int main()
{
    int n, i, num, count = 0;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    int arr[n];

    printf("Enter %d integers:\n", n);
    for (i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    printf("Enter the number to find its frequency: ");
    scanf("%d", &num);

    for (i = 0; i < n; i++)
    {
        if (arr[i] == num)
        {
            count++;
        }
    }

    printf("The frequency of %d is: %d\n", num, count);

    return 0;
}
```

```
PS C:\Users\sanch\OneDrive\Desktop\c programming> cd 'c:\Users\sanch\OneDrive\Desktop\c programming\exp_4.c\output'
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_4.c\output> & .\'exp3.exe'

Enter the number of elements: 10
Enter 10 integers:
1 76 0 -23 0 11 1 64 0 -78
Enter the number to find its frequency: 1
The frequency of 1 is: 2
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_4.c\output>
```

Exp = 4.4

→

WAP that reads two matrices and computes product. Print both input matrices and resultant matrix with suitable headings & output should be in matrix format only. Program must check the compatibility of orders of matrices for multiplication. Report appropriate message in case of incompatibility

→

```
#include <stdio.h>

int main ()
{
int m, n, p, q, i, j, k;

printf (" Enter rows & columns of matrix A:")
scanf ("%d %d", &m, &n);

int A [m] [n];
printf (" Enter elements of matrix A (row-wise):\n")
```

```
for (i = 0; i < m; i++)
 {
    for (j = 0; j < n; j++)
      {
         scanf("%d", &A[i][j]);
      }
 }

printf("Enter rows & columns of matrix B : ");
 scanf("%d %d", &p, &q);

int B [p][q];
printf("Enter elements of matrix B (row wise) :\n");
 for (i = 0; i < p; i++)
  {
     for (j = 0; j < q; j++)
      {
         scanf("%d", &B[i][j]);
      }
  }

if (n != p)
{
   printf("\n Matrix multiplication not possible \n");
   printf("Reason: columns of A (%d) not equal to
```

```
Row of (%d)\n", n, p);
 return 0;
}

int C[m][q]
 for (i=0, i<m; i++)
  {
     for (j=0; j<q; j++)
     {
         c[i][j] = 0;
         for (K=0; K<n; K++)
         {
            C[i][j] = C[i][j] + A[i][k] * B[K][j];
         }
     }
  }

printf ("\n matrix A: \n");
  for (i=0; i<m; i++)
   {
      for (j=0; j<n; j++)
       {
           printf ("%d", A[i][j]);
       }
      printf ("\n");
   }
```

```
printf("\n Matrix B :\n");
  for(i=0; i<p; i++)
 {
   for(j=0; j<q; j++)
    {
      printf("%4d", B[i][j]);
    }
    printf("\n");
 }

printf("\n Resultant Matrix (AxB): \n");
  for(i=0; i<m; i++)
  {
    for(j=0; j<q; j++)
      {
        printf("%4d", C[i][j]);
      }
     printf("\n");
  }

  return 0;
}
```

```c
#include <stdio.h>

int main()
{
    int m, n, p, q, i, j, k;

    printf("Enter rows and columns of Matrix A: ");
    scanf("%d %d", &m, &n);

    int A[m][n];
    printf("Enter elements of Matrix A (row-wise):\n");
    for (i = 0; i < m; i++)
    {
        for (j = 0; j < n; j++)
        {
            scanf("%d", &A[i][j]);
        }
    }

    printf("Enter rows and columns of Matrix B: ");
    scanf("%d %d", &p, &q);

    int B[p][q];
    printf("Enter elements of Matrix B (row-wise):\n");
    for (i = 0; i < p; i++)
    {
        for (j = 0; j < q; j++)
        {
            scanf("%d", &B[i][j]);
        }
    }

    if (n != p)
    {
        printf("\nMatrix multiplication not possible!\n");
        printf("Reason: Columns of A (%d) not equal to Rows of B (%d)\n", n, p);
        return 0;
    }

    int C[m][q];
    for (i = 0; i < m; i++)
    {
        for (j = 0; j < q; j++)
        {
            C[i][j] = 0;
            for (k = 0; k < n; k++)
            {
                C[i][j] = C[i][j] + A[i][k] * B[k][j];
            }
        }
    }

    printf("\nMatrix A:\n");
    for (i = 0; i < m; i++)
    {
        for (j = 0; j < n; j++)
        {
            printf("%4d", A[i][j]);
        }
        printf("\n");
    }

    printf("\nMatrix B:\n");
    for (i = 0; i < p; i++)
    {
        for (j = 0; j < q; j++)
        {
            printf("%4d", B[i][j]);
        }
        printf("\n");
    }

    printf("\nResultant Matrix (A x B):\n");
    for (i = 0; i < m; i++)
    {
        for (j = 0; j < q; j++)
        {
            printf("%4d", C[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

```
PS C:\Users\sanch\OneDrive\Desktop\c programming> cd 'c:\Users\sanch\OneDrive\Desktop\c programming\exp_4.c\output'
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_4.c\output> & .\'exp4.exe'
Enter rows and columns of Matrix A: 2 3
Enter elements of Matrix A (row-wise):
1 2 3 4 5 6
Enter rows and columns of Matrix B: 3 2
Enter elements of Matrix B (row-wise):
7 8 9 10 11 12

Matrix A:
   1   2   3
   4   5   6

Matrix B:
   7   8
   9  10
  11  12

Resultant Matrix (A x B):
  58  64
 139 154
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_4.c\output>
```